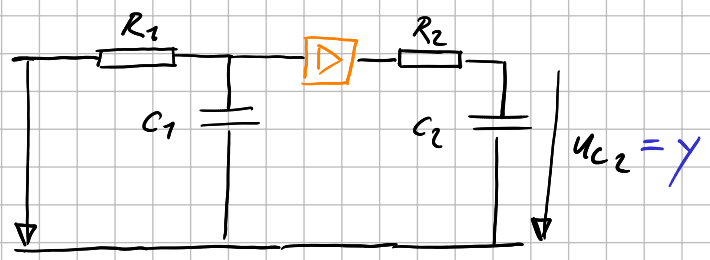
Ersatzschaltbild

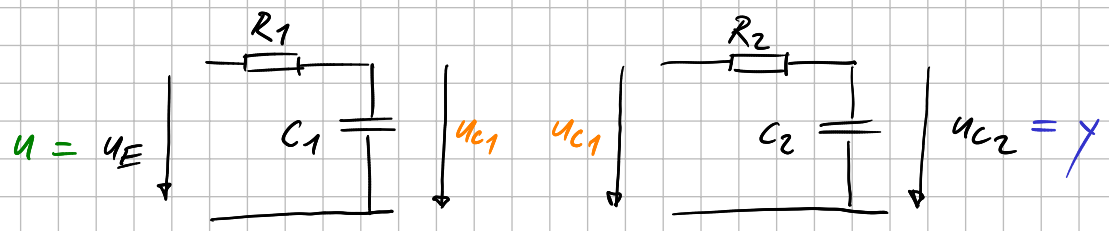
System Eingang $u = u_E$



System Ausgang

Impedanzwandler / Spannungsfolger

- Unterbricht Stromfluss
- Spannung am Ausgang entspricht Spannung am Eingang



1 DGL

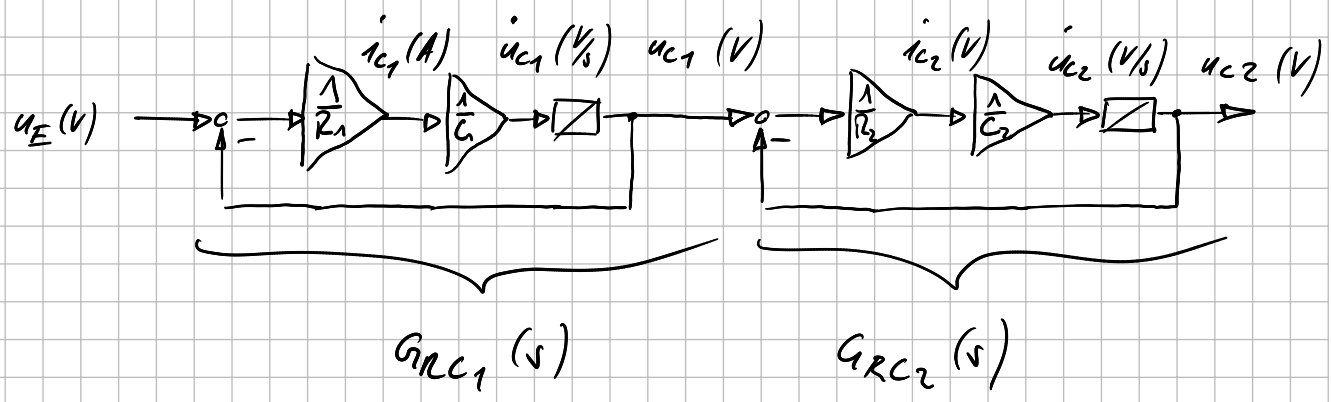
$$u_E = R_1 C_1 \dot{u}_{C1} + u_{C1} \qquad (PT_1)$$

$$u_{C1} = R_2 C_2 \dot{u}_{C2} + u_{C2} \qquad (PT_1)$$

2 Blockschaltbild

$$i_{C1} = \frac{1}{R_1}(u_E - u_{C1}) \qquad \dot{u}_{C1} = \frac{1}{C_1} i_{C1}$$

$$i_{C2} = \frac{1}{R_2}(u_{C1} - u_{C2}) \qquad \dot{u}_{C2} = \frac{1}{C_2} i_{C2}$$



$$G(s) = \frac{1}{R_1C_1s+1} \cdot \frac{1}{R_2C_2s+1}$$

(PT_2) (PT_1) (PT_1)

3 Übertragungsfunktion

$$R_1 = R_2 = R, \quad C_1 = C_2 = C$$

$$G(s) = \frac{1}{(\underbrace{RC}_{T_1}s+1)^2} \quad \text{Zeitkonstantenform}$$

$$T_1 = RC = 2{,}2\,\text{msec} \quad //$$

4 Frequenzgang

$$G(\omega) = G(s)\Big|_{s=j\omega} = \frac{1}{(RCj\omega+1)^2}$$

$$\omega_{g1,2} = \frac{1}{T_1} = \frac{1}{RC} = 452{,}7\ \text{rad/sec} \quad //$$

5

Statische Verstärkung

$$G(0) = 1 \quad //$$

Null- und Polstellen

$$z = \{\} \qquad p_{1,2} = -\frac{1}{T_1} = -\frac{1}{RC} = -452{,}7 \quad //$$

6 Sprungantwort PT_2 mit Laplace

- System startet aus Ruhelage
  → verschwindende Anfangsbedingungen

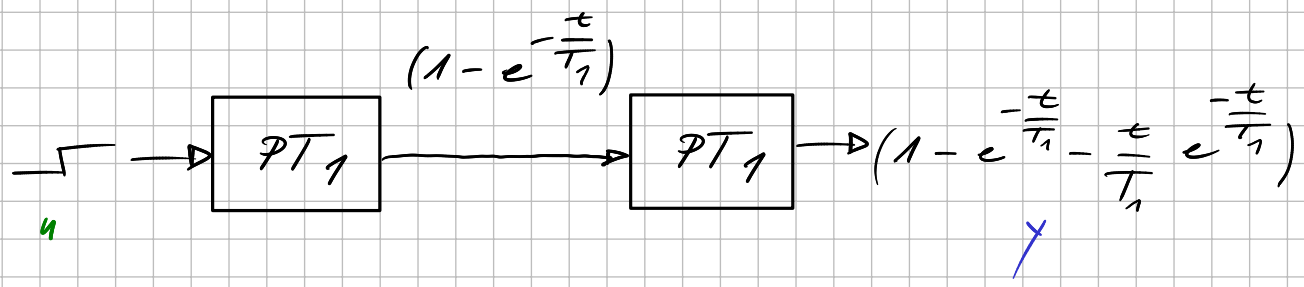
$$Y(s) = G(s)\,\underbrace{U(s)}_{\frac{1}{s}}$$

$$\frac{1}{s} \circ\!\!-\!\!\circ\; u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & \text{sonst} \end{cases}$$

$$Y(s) = \frac{1}{(T_1 s + 1)^2}\,\frac{1}{s}$$

$$y(t) = 1 - e^{-\frac{t}{T_1}} - \frac{t}{T_1}\,e^{-\frac{t}{T_1}} \;//$$

7 Antwort PT_1 auf Sprungantwort PT_1



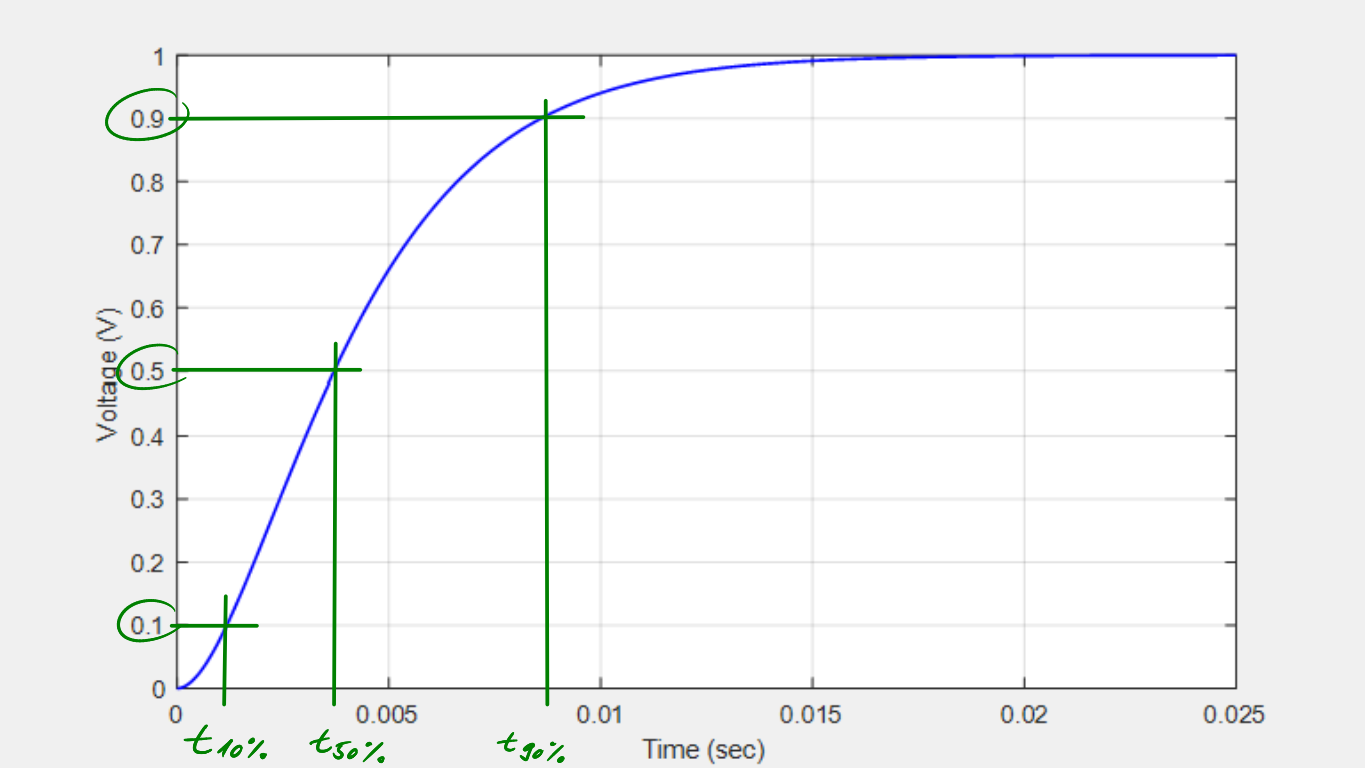
8 Steigung Sprungantwort bei $t=0$

$$\dot{y}(t) = \cancel{\frac{1}{T_1}\,e^{-\frac{t}{T_1}}} - \cancel{\frac{1}{T_1}\,e^{-\frac{t}{T_1}}} + \frac{t^2}{T_1^2}\,e^{-\frac{t}{T_1}}$$

$$= 0 \;//$$
$$\Big|_{t=0}$$

9 & 10 Zeit-Prozent-Kennwerte für PT_2

Mit Matlab ausgewertet für $t = (0:1e\text{-}6:0{,}025)^T$



$t_{10\%} = 1{,}2$ msec

$t_{50\%} = 3{,}7$ msec

$t_{90\%} = 8{,}6$ msec //

$t_{10\%}/t_{50\%} = 0{,}32$

$t_{10\%}/t_{90\%} = 0{,}14$ //

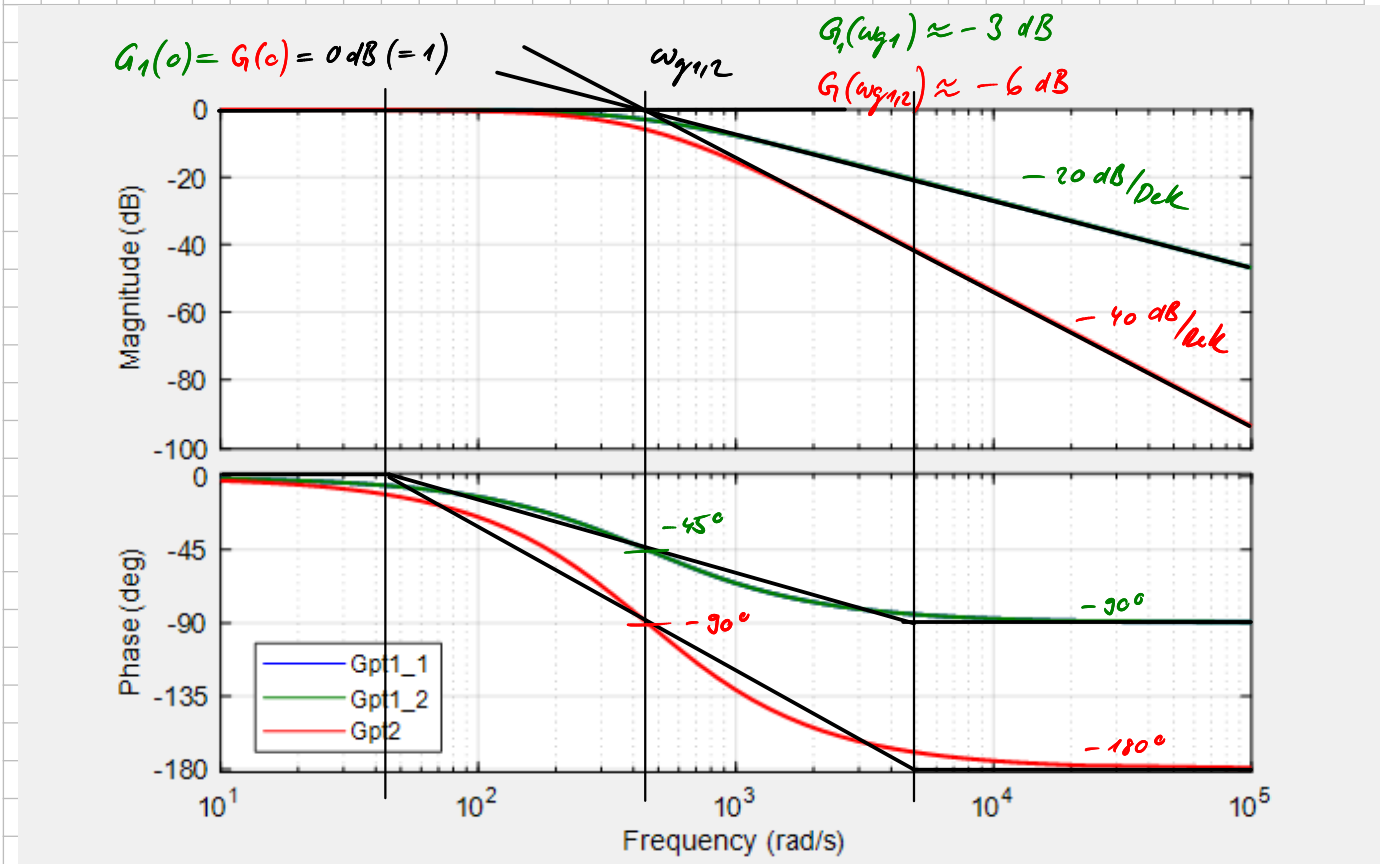
$t_{10\%}/T_1 = 0{,}53 = \alpha_{10\%}$

$t_{50\%}/T_1 = 1{,}69 = \alpha_{50\%}$

$t_{90\%}/T_1 = 3{,}89 = \alpha_{90\%}$ //

11

$$\boxed{T_1 = \frac{1}{3}\left(\frac{t_{10\%}}{\alpha_{10\%}} + \frac{t_{50\%}}{\alpha_{50\%}} + \frac{t_{90\%}}{\alpha_{90\%}}\right)}$$

12

$$G_1(s) = G_2(s) = \frac{1}{\frac{1}{\omega_g} s + 1} \qquad G(s) = \frac{1}{\left(\frac{1}{\omega_g} s + 1\right)^2} = G_1(s) \cdot G_2(s)$$